Five number summary

1

a

i $16.12, 16.83, 19.44, 21.88, 22.95, 23.62, 23.74, 24.19, 24.41, 24.52, 25.64$

ii 25.64

iii 16.12

iv 23.62

v 19.44

vi 24.41

b

i $51, 53, 55, 55, 63, 66, 68, 70, 75, 81, 84, 87, 89, 90, 92$

ii 92

iii 51

iv 70

v 55

vi 87

c

i $53, 58, 59, 62, 62, 64, 75, 77, 79, 80, 81, 82, 86, 96$

ii 96

iii 53

iv 76

v 62

vi 81

2

a 114 **b** 119.5 **c** 104

3

a **b** 50% **c** No

Minimum	18
Q1	28
Median	38
Q3	48
Maximum	63

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4

a $27\frac{1}{2}$ minutes **b** 45 minutes **c** 10 minutes **d** 35

e 83% **f** 22%

5 20

6 Between 11.5 and 18.

7 8.8

8

a 142 minutes **b** 142 minutes **c** 27 **d** 137 minutes

9

a 18.97 kg **b**

Q1	17 kg
Median	19 kg
Q3	21 kg

c 2nd

Box-and-whisker plots

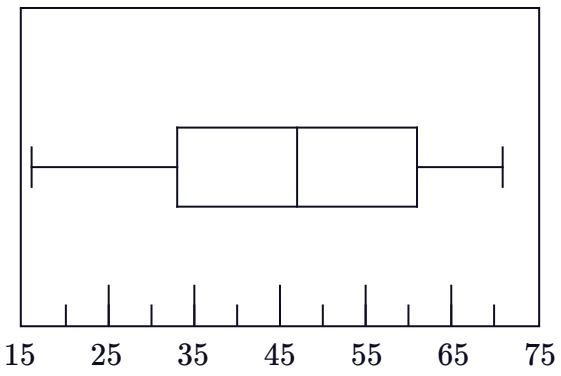
10

a 3 **b** 18 **c** 15 **d** 10

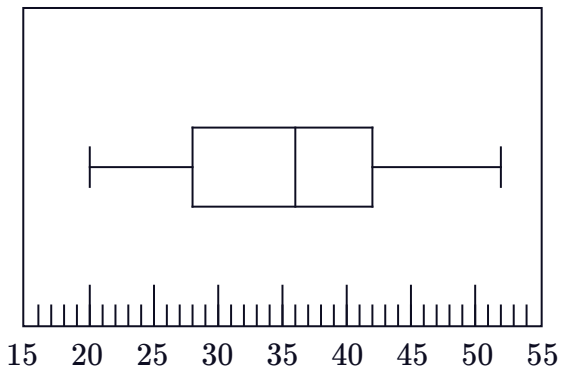
e 7

11

a

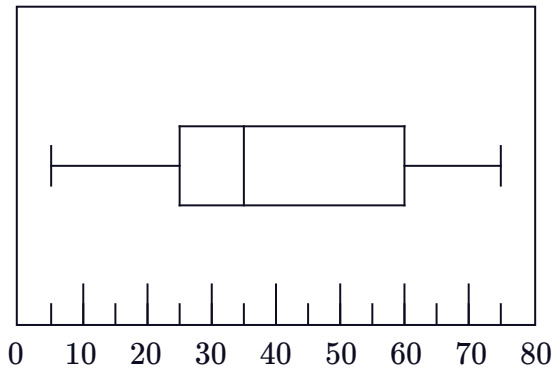


b

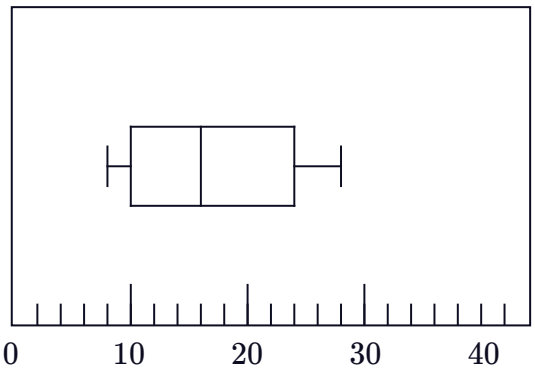


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c



12

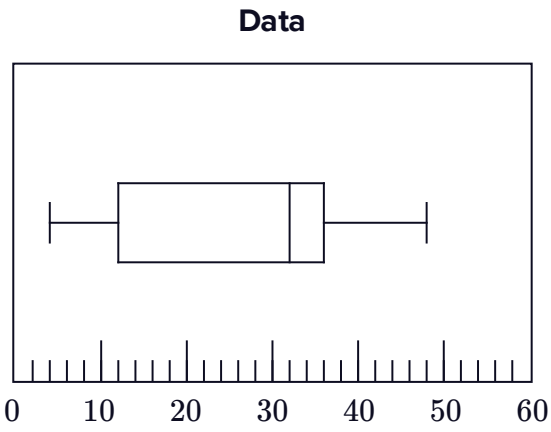


13

a

Minimum	4
Q1	12
Median	32
Q3	36
Maximum	48

b



14

a 31 years **b** 17 years **c** 25% **d** 7

e 1st **f** 5

15

a

i 50% **ii** 25% **iii** 50% **iv** 75%

v 75%

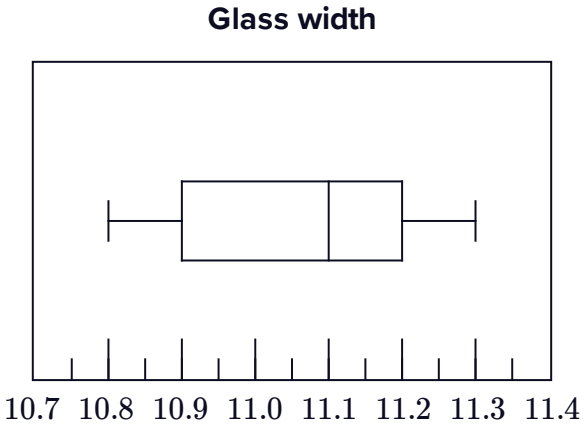
b 2nd

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a 11.10 mm

b 0.3 mm

c



d 78.95%

e 2nd

f The mode and mean

17

a 50 beats per minute

b 62 beats per minute

c Athletes

d 14

e 24

f Athletes

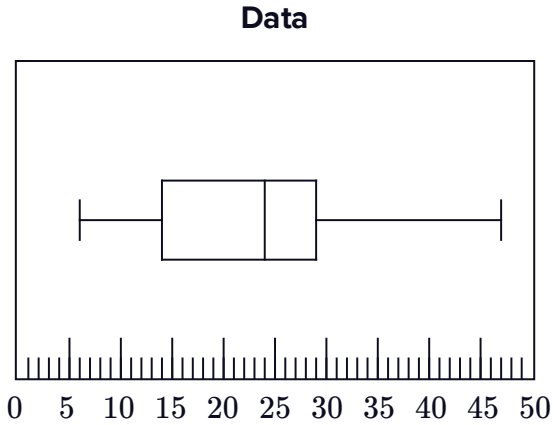
18 Although for at least 75% of the time, he blacks out for less than 4.5 seconds, he will occasionally black out for longer than 7 seconds.

Box-and-whisker plots with technology

19

a

i



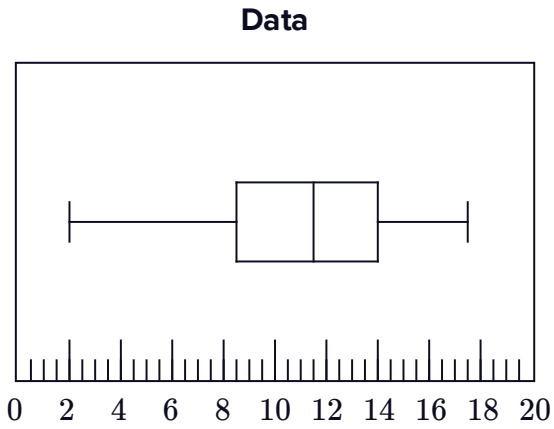
ii

Minimum	6
Q1	14
Median	24
Q3	29
Maximum	47

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b

i

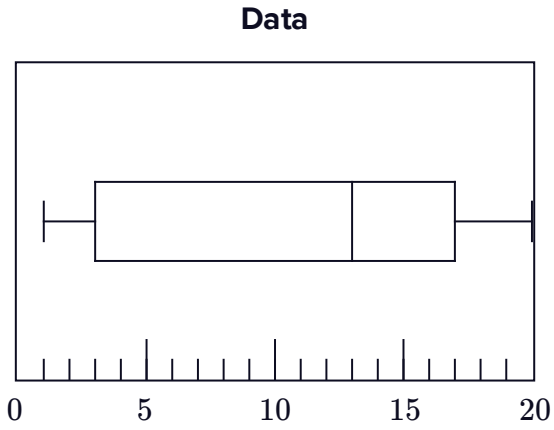


ii

Minimum	2
Q1	8.5
Median	11.5
Q3	14
Maximum	17.5

c

i

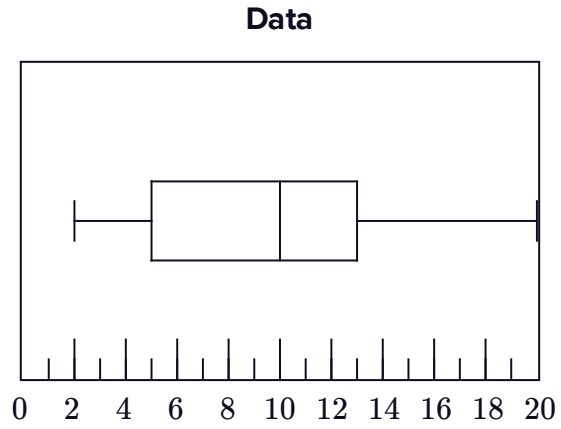


ii

Minimum	1
Q1	3
Median	13
Q3	17
Maximum	20

20

a



b Mode

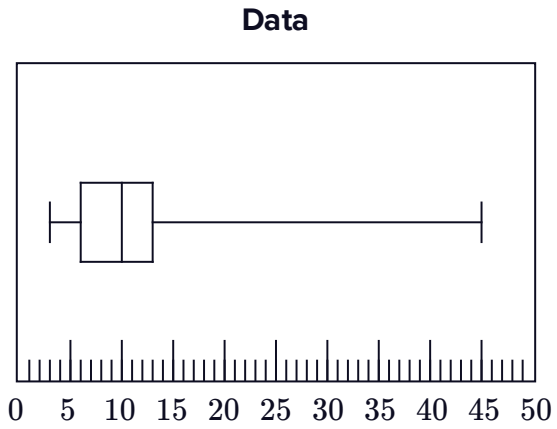
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c

Minimum	2
Q1	5
Median	10
Q3	13
Maximum	20

21

a



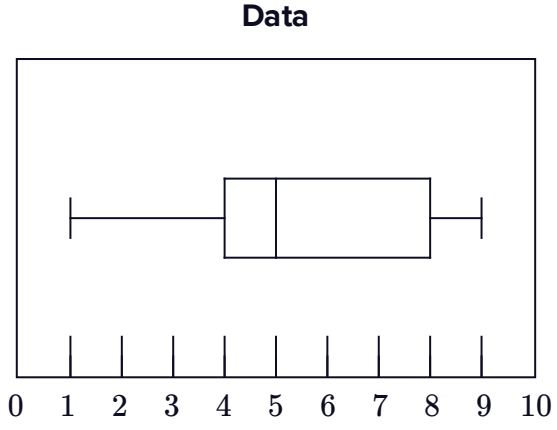
b Outlier

c

Minimum	3
Q1	6
Median	10
Q3	13
Maximum	45

22

a



b Mode

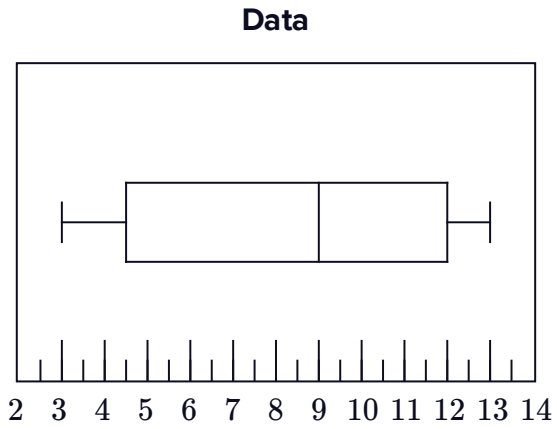
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c

Minimum	1
Q1	4
Median	5
Q3	8
Maximum	9

23

a



b No

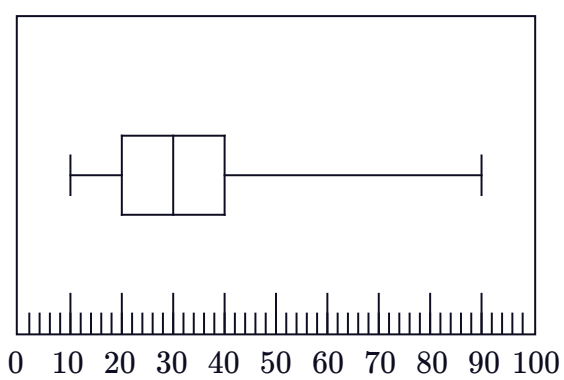
c

Minimum	3
Q1	4.5
Median	9
Q3	12
Maximum	13

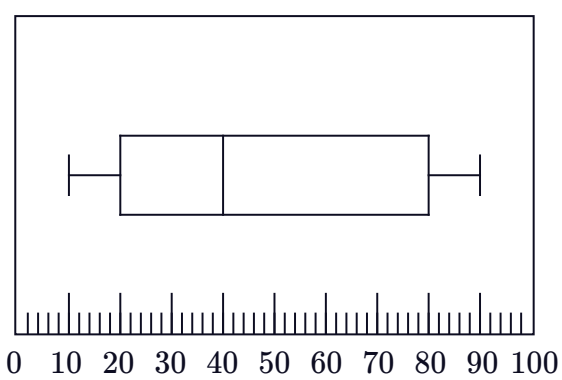
Box-and-whisker plots and the shape of data

24

a

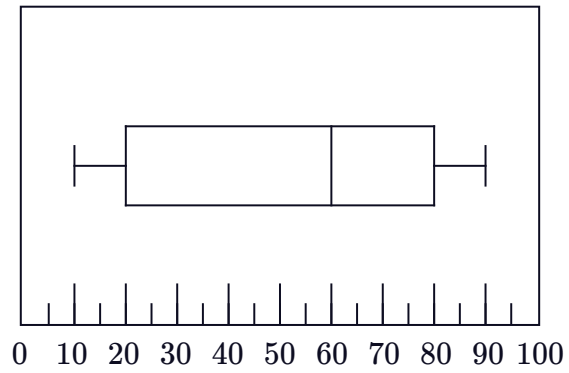


b

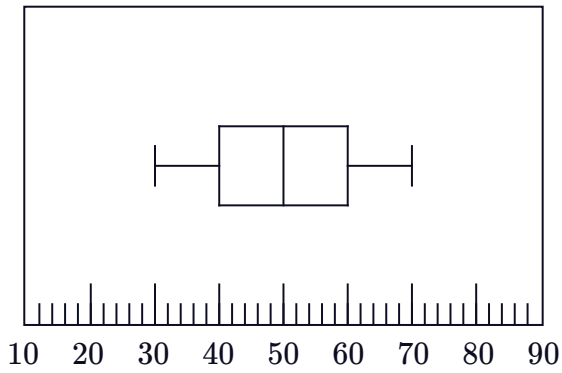


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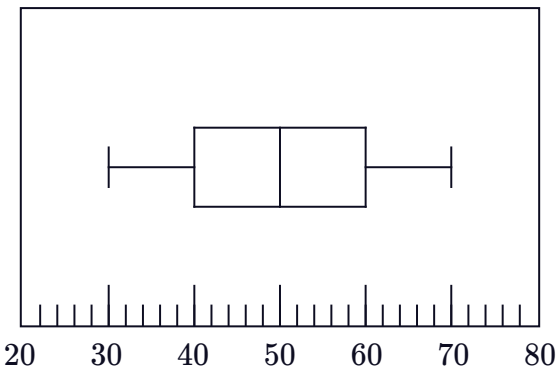
c



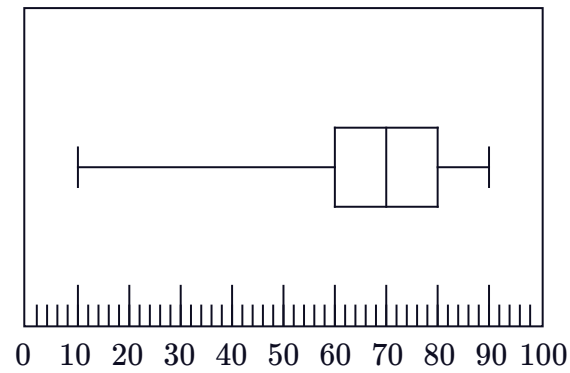
d



e



f



25

- Box Plot 1/Histogram A
- Box Plot 2/Histogram B
- Box Plot 3/Histogram D
- Box Plot 4/Histogram C

26

a Yes **b** No **c** No **d** Yes

e Yes **f** No

27

a The box-and-whisker plot has a long tail to the right which indicates positive skew, while the histogram does not appear to be skewed. The median for the histogram is roughly in the middle, while the median of the box-and-whisker plot is located further to the left.

b The box-and-whisker plot indicates that the data is widely spread, while the histogram indicates that most of the data occurs near the median.

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